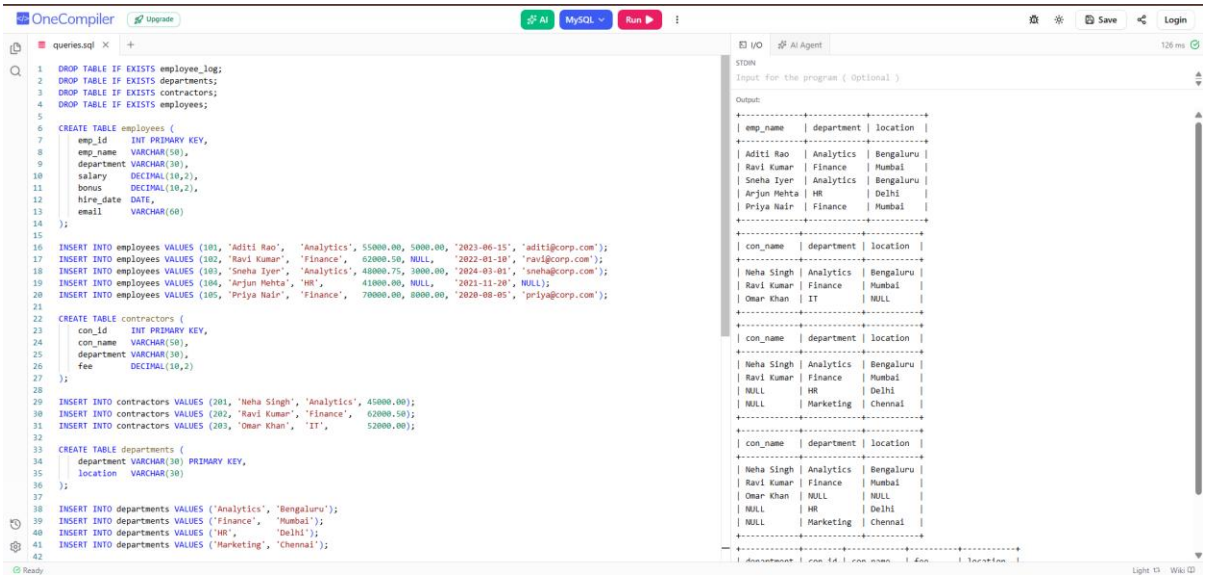
DATABASE MANAGEMENT SYSTEM LAB

Name: Aditi M
USN: 1RUA25BCA0006
Subject: Database Management System

LAB 7

Question: Implement different types of Joins: Implement various types of joins in SQL, including Inner Join, Outer Join, and Natural Join, with suitable examples.

Output:



```
DROP TABLE IF EXISTS employee_log;
DROP TABLE IF EXISTS departments;
DROP TABLE IF EXISTS contractors;
DROP TABLE IF EXISTS employees;

CREATE TABLE employees (
    emp_id     INT PRIMARY KEY,
    emp_name   VARCHAR(50),
    department VARCHAR(30),
    salary     DECIMAL(10,2),
    bonus      DECIMAL(10,2),
    hire_date  DATE,
    email      VARCHAR(60)
);

INSERT INTO employees VALUES (101, 'Aditi Rao',   'Analytics', 55000.00, 5000.00, '2023-06-15', 'aditi@corp.com');
INSERT INTO employees VALUES (102, 'Ravi Kumar',  'Finance',   62000.50, NULL,    '2022-01-10', 'ravi@corp.com');
INSERT INTO employees VALUES (103, 'Sneha Iyer',  'Analytics', 48000.75, 3000.00, '2024-03-01', 'sneha@corp.com');
INSERT INTO employees VALUES (104, 'Arjun Mehta', 'HR',        41000.00, NULL,    '2021-11-20', NULL);
INSERT INTO employees VALUES (105, 'Priya Nair',  'Finance',   70000.00, 8000.00, '2020-08-05', 'priya@corp.com');

CREATE TABLE contractors (
    con_id     INT PRIMARY KEY,
    con_name   VARCHAR(50),
    department VARCHAR(30),
    fee        DECIMAL(10,2)
);

INSERT INTO contractors VALUES (201, 'Neha Singh', 'Analytics', 45000.00);
INSERT INTO contractors VALUES (202, 'Ravi Kumar', 'Finance',   62000.50);
INSERT INTO contractors VALUES (203, 'Omar Khan',  'IT',        52000.00);

CREATE TABLE departments (
    department VARCHAR(30) PRIMARY KEY,
    location   VARCHAR(30)
);

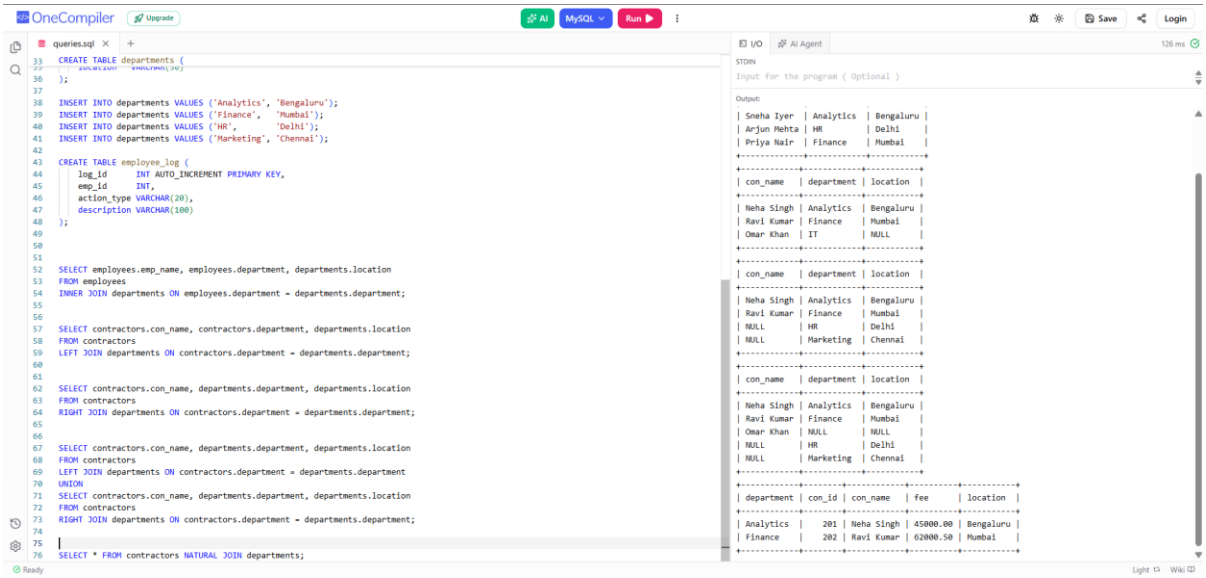
INSERT INTO departments VALUES ('Analytics', 'Bengaluru');
INSERT INTO departments VALUES ('Finance',   'Mumbai');
INSERT INTO departments VALUES ('HR',        'Delhi');
INSERT INTO departments VALUES ('Marketing', 'Chennai');
```

```
+------------+------------+----------+
| emp_name   | department | location |
+------------+------------+----------+
| Aditi Rao  | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| Sneha Iyer | Analytics  | Bengaluru |
| Arjun Mehta| HR         | Delhi    |
| Priya Nair | Finance    | Mumbai   |
+------------+------------+----------+

+----------+------------+----------+
| con_name | department | location |
+----------+------------+----------+
| Neha Singh | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| Omar Khan  | IT         | NULL     |
+----------+------------+----------+

+----------+------------+----------+
| con_name | department | location |
+----------+------------+----------+
| Neha Singh | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| NULL       | HR         | Delhi    |
| NULL       | Marketing  | Chennai  |
+----------+------------+----------+

+----------+------------+----------+
| con_name | department | location |
+----------+------------+----------+
| Neha Singh | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| Omar Khan  | NULL       | NULL     |
| NULL       | HR         | Delhi    |
| NULL       | Marketing  | Chennai  |
+----------+------------+----------+
```



```
CREATE TABLE departments (
);

INSERT INTO departments VALUES ('Analytics', 'Bengaluru');
INSERT INTO departments VALUES ('Finance',   'Mumbai');
INSERT INTO departments VALUES ('HR',        'Delhi');
INSERT INTO departments VALUES ('Marketing', 'Chennai');

CREATE TABLE employee_log (
    log_id      INT AUTO_INCREMENT PRIMARY KEY,
    emp_id      INT,
    action_type VARCHAR(20),
    description VARCHAR(100)
);



SELECT employees.emp_name, employees.department, departments.location
FROM employees
INNER JOIN departments ON employees.department = departments.department;


SELECT contractors.con_name, contractors.department, departments.location
FROM contractors
LEFT JOIN departments ON contractors.department = departments.department;


SELECT contractors.con_name, departments.department, departments.location
FROM contractors
RIGHT JOIN departments ON contractors.department = departments.department;


SELECT contractors.con_name, departments.department, departments.location
FROM contractors
LEFT JOIN departments ON contractors.department = departments.department
UNION
SELECT contractors.con_name, departments.department, departments.location
FROM contractors
RIGHT JOIN departments ON contractors.department = departments.department;

SELECT * FROM contractors NATURAL JOIN departments;
```

```
| Sneha Iyer | Analytics  | Bengaluru |
| Arjun Mehta| HR         | Delhi    |
| Priya Nair | Finance    | Mumbai   |
+------------+------------+----------+

+----------+------------+----------+
| con_name | department | location |
+----------+------------+----------+
| Neha Singh | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| Omar Khan  | IT         | NULL     |
+----------+------------+----------+

+----------+------------+----------+
| con_name | department | location |
+----------+------------+----------+
| Neha Singh | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| NULL       | HR         | Delhi    |
| NULL       | Marketing  | Chennai  |
+----------+------------+----------+

+----------+------------+----------+
| con_name | department | location |
+----------+------------+----------+
| Neha Singh | Analytics  | Bengaluru |
| Ravi Kumar | Finance    | Mumbai   |
| Omar Khan  | NULL       | NULL     |
| NULL       | HR         | Delhi    |
| NULL       | Marketing  | Chennai  |
+----------+------------+----------+

+------------+--------+----------+---------+----------+
| department | con_id | con_name | fee     | location |
+------------+--------+----------+---------+----------+
| Analytics  |    201 | Neha Singh | 45000.00 | Bengaluru |
| Finance    |    202 | Ravi Kumar | 62000.50 | Mumbai   |
+------------+--------+----------+---------+----------+
```